Haoquan Wang

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EDUCATION

University of Utah Salt Lake City,Utah | Jan 2025 - Excepted May 2028

Bachelor of Science in Computer Science (Minor: Mathematics) | GPA: 3.857 (49.0 Credits)
Relevant Coursework**:**
Linear Algebra (A) | Discrete Mathematics (A) | Data Structures (A-) | Calculus I & II (A)
Academic Awards: Dean's List, Fall 2025 – Present | Utah Global Scholarship, Spring 2025 - Present
Skills:
Languages: Python, Java, C/C++
Tools: Pandas, PyTorch, TorchVision, NumPy, Mbed OS, Git, JUnit, SHAP

WORK

Wuhan Changhang Keda Engineering | Office Staff August 2022 - October 2023

- Refined multi-dimensional statistical models, improving joint-distribution accuracy by 3%.
- Analyzed complex data trends to deliver key insights, mitigating compliance risks by 7%.

PROJECTS

Research Paper: High-Order Temporal Graph Learning for Network Inference | Python, PyTorch September 2025 - present

- Co-authored working paper proposing a novel high-order temporal graph neural network (HTGN) to infer malicious anomalies in evolving enterprise networks.
- Formulated a permutation-invariant framework and designed a smooth Softplus attention mechanism to suppress noise.
- Evaluated model on the LANL dataset using an RTX 4090, outperforming strong baselines across AUC, AP, and recall.

Explainable AI (XAI) & Model Robustness Benchmark | Python, PyTorch SHAP LIME
June 2026 - present

- Integrating foundational feature-attribution tools (e.g., SHAP, LIME) to extract and visualize feature importance, enhancing the interpretability of black-box model decisions

Deep Learning: Image Classification with Data Augmentation | Python, PyTorch, CUDA
August 2025 - October 2025

- Built a CIFAR-10 classification pipeline, achieving a 4% accuracy boost (87% to 91%) and reducing the overfitting gap to near-zero.
- Applied multiple data augmentation techniques (RandomHorizontalFlip, RandomCrop, Engineered a GPU-accelerated training pipeline using CUDA, optimizing batch processing to deliver a 10x-20x speedup over CPU.

Autonomous Path-Finding Drone | C++, Arm Mbed OS July 2023 - March 2024

- Engineered a navigation drone using C++ on the Arm Mbed OS platform, implementing real-time sensor data processing for autonomous flight.
- Optimized control algorithms to bridge the gap between high-level path planning and low-level motor execution, significantly improving flight stability.